



Beyond happy and sad: Exploring granular affect labeling to enhance emotion regulation ability

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HIGHLIGHTS

- We introduce granular affect labeling as a theory-informed approach for proactively supporting emotion regulation.
- We design and implement a conversational agent grounded in the Theory of Constructed Emotions to scaffold the development of granular emotion concepts.
- A 24-day field study demonstrates that granular affect labeling significantly improves emotion regulation ability without explicitly teaching regulation strategies.
- Qualitative findings reveal how expanded emotion vocabularies support emotion-cause understanding and reflective regulation.

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ABSTRACT

Growing evidence in affective neuroscience suggests that emotion words do more than simply label emotions; they actively shape emotion concepts, which in turn influence how we experience and regulate emotions. Emotion concepts are mental representations that the brain uses to categorize and make sense of affective experiences. Expanding one's repertoire of nuanced emotion concepts has been linked to better emotion regulation, yet this promising potential has been largely overlooked by existing technologies and interventions. To address this, we developed a conversational agent informed by the Theory of Constructed Emotions, employing *granular affect labeling* to expand users' emotion concepts and strengthen regulation abilities. Through a 24-day controlled field study involving 33 participants, we found that our chatbot broadens participants' active emotion vocabulary—a direct window to one's emotion concepts—and significantly improves their ability to regulate emotions compared to a control emotional support chatbot. Our findings contribute to a deeper understanding of how granular affect labeling can be leveraged in technology to cultivate more nuanced emotional experiences and enhance well-being.

1. Introduction

What if the words we use to describe our emotions are not merely labels but actively shape our emotional reality? While this might sound like a philosophical reflection, recent neuroscience research reveals a startling truth: the language we use to describe our feelings profoundly structures our emotion concepts and alters how we experience and regulate them (Lindquist et al., 2015b).

Far from being mere labels, words are deeply connected to mental concepts in the brain. Empirical evidence from neuroscience demonstrates that words activate brain regions associated with the concepts they represent (Pulvermüller, 2005). For instance, the word “kick”

engages areas of the brain linked to the leg and the action of kicking, while “run” or “grasp” activates motor regions associated with those actions (Hauk et al., 2004). This phenomenon, known as embodied cognition, demonstrates the interconnection between words and sensory-motor systems. Likewise, emotion words are observed to be deeply linked to emotion concepts. For example, hearing words such as “angry” or “happy” activates distributed brain regions beyond linguistic processing (Lee and Satpute, 2024). When the semantic meaning of emotion words is impaired,¹ emotion perception itself becomes impaired (Lindquist et al., 2006, 2014; Gendron et al., 2012). In short, the

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¹ The semantic meaning of a word can be impaired by repeating the word out loud many times (also known as the semantic satiation technique).

presence of emotion words activates brain regions associated with corresponding emotional processes, while their absence impairs the ability to perceive emotions.

The above findings suggest that emotional language is not merely for describing pre-existing emotions but plays a constitutive role in shaping the formation of emotions themselves (Lee and Satpute, 2024; Lindquist et al., 2006, 2014; Gendron et al., 2012). Modern affective science supports this perspective, proposing that emotions are learned mental concepts, with language playing a crucial role in the learning process (Hoemann et al., 2019; Barrett, 2017b; Lindquist et al., 2015b). The brain relies on these concepts to regulate bodily states (Barrett, 2017b). Therefore, the granularity of one's emotional vocabulary directly impacts the brain's ability to manage emotions. Consider how differently you might experience stress if you can only label it as "bad" versus recognizing it as "overwhelming but motivating" or "intense yet temporary." This level of granularity in emotional labeling is not just linguistic sophistication—it reflects fundamentally different ways of experiencing and managing emotions.

The practical implications of emotional granularity are significant. People who label their emotions with great precision and make distinctions between emotions like "annoyed" and "enraged" instead of simply labeling them as "bad" tend to regulate their emotions more effectively (Lee et al., 2017; Erbas et al., 2022; Tan et al., 2022a). They are also less prone to drug and alcohol abuse and enjoy better physical health (Kashdan et al., 2010a). Using words to label emotions, known as affect labeling, has shown promising results. For instance, individuals with arachnophobia experienced less anxiety and were more willing to approach spiders when trained to articulate their emotions compared to those using cognitive reappraisal or distraction techniques (Lieberman et al., 2011).

Although the significance of granular emotion concepts for regulation is evident, designing technology to support their development presents unique challenges. Emotions are inherently fleeting experiences influenced by momentary bodily sensations, contextual cues, and personal past experiences (Barrett, 2017a). Prior works in emotion regulation support technologies (ERST) primarily focus on *reactive* interventions—helping users track, express, reappraise, or regulate emotions after they arise (Slovak et al., 2023; Sadka and Antle, 2022). Although valuable, these approaches do not address the foundational need to develop richer emotional concepts that could enhance regulatory capacity. Addressing this gap, we introduce a novel approach called granular affect labeling, which supports the proactive development of emotional granularity and enhances emotion regulation.

We draw from the Theory of Constructed Emotions (TCE) (Barrett, 2017b), which explains how emotions develop and function. Based on the TCE's theoretical principles, we derive design guidelines and develop a chatbot that facilitates real-time, guided, and reflective affect labeling, enabling users to construct granular emotion concepts. Through a 24-day field study with 33 participants, we found that our chatbot significantly improved users' emotion regulation abilities, while a control chatbot designed solely to provide emotional support did not. These findings demonstrate the effectiveness of granular affect labeling. In summary, our study contribution is threefold:

- We present the design and implementation of a chatbot that supports the development of granular emotion concepts, following four design guidelines we derived from the TCE.
- We present quantitative results from a field study showing that granular affect labeling significantly improves emotion regulation abilities.
- We provide qualitative findings that offer new insights into how a granular affect labeling chatbot helps people improve their emotion regulation abilities.

2. Theoretical background and related work

Our work is grounded in the Theory of Constructed Emotions (TCE) and builds on existing digital approaches that develop emotional understanding and regulatory skills through language. The TCE views emotions as concepts that are actively constructed through experience rather than innately given, indicating an opportunity for technological scaffolding of this learning process. It also provides a clear mechanism for how granular emotion concepts contribute to improved regulation, making it a suitable framework to derive design guidelines (Taveter et al., 2019; Taveter and Iqbal, 2021; Taveter and Kirikal, 2022). According to the TCE, emotion formation is deeply connected to the brain's energy regulation role. To fully understand how emotions are abstractive mental concepts but at the same time result in tangible physical experiences, it is essential to examine the brain's functioning and understand how emotions emerge as embodied mental constructs. Accordingly, the following subsections outline the brain's operational mechanisms, detail the constituent components of emotion emergence, review the relevant literature, and highlight how our work extends current research.

2.1. The theory of constructed emotion

The predictive brain

The brain's main function is to regulate energy to ensure the organism's survival and reproduction (Barrett, 2017b). In complex body systems such as the human body, this regulation is achieved through *allostasis* (Sterling, 2012; Schulkin and Sterling, 2019)—a process of proactively anticipating physiological needs and preparing to meet them before they arise. To perform allostasis effectively, the brain employs predictive processing: rather than passively reacting to stimuli, it proactively predicts upcoming sensory events — both internal and external — and prepares appropriate responses (Friston, 2010; Clark, 2013; Hohwy and Seth, 2020). Central to this predictive system is *interoception*: the brain's moment-to-moment modeling of the internal physiological state of the body (Craig, 2002; Sterling, 2012; Sennesh et al., 2022). These signals are high dimensional and come from the various organs in the body. Interoception relies on comparing a predictive model to incoming sensory input arising within the body (Barrett and Quigley, 2021). While much of interoceptive processing occurs outside of conscious awareness, it manifests in consciousness as affect—a low-dimensional representation of the body's state characterized by valence (pleasant/unpleasant) and arousal (high/low energy) (Barrett and Bliss-Moreau, 2009). These affective states are experienced as bodily sensations, providing a conscious window into the body's ongoing physiological conditions.

Emotions are embodied mental concepts

To effectively regulate the body, the brain must not only anticipate incoming signals but also derive their meaning and determine the best course of action (Barrett, 2017a). The TCE proposes that the brain accomplishes this by comparing incoming signals to known *concepts* (Barrett, 2017b). Concepts are the abstractive mental representations that reflect knowledge and expectations of body sensations, external context such as environmental cues, events, causes, consequences, and more (Barrett et al., 2007b). According to TCE, emotions are concepts that the brain generates as part of its regulatory role in maintaining bodily functions. For example, a churning stomach at lunchtime is likely to match the hunger concept, prompting the brain to prepare for digestion. In contrast, the same sensation before a presentation will best suit the anxiety concept, triggering the release of stress hormones and preparing the body for a fight-or-flight response. Based on this working mechanism, the TCE proposes that emotions are not fixed but are (1) learned over time and (2) dynamically constructed based on internal sensations, external context, and past experiences (Barrett, 2014).

The role of language in emotion learning

Words are essential in the creation of concepts, especially abstract ones (Lupyan et al., 2007; Borghi et al., 2018; Henningsen-Schomers et al., 2022). Works in cognitive science have found that assigning words to label a set of objects of unrelated functionality facilitates the categorization process (Waxman and Markow, 1995; Lupyan et al., 2007; Hoemann et al., 2021). Using neurobiologically constrained neural networks, Henningsen-Schomers et al. (2022) demonstrated that verbal labels significantly improved concept formation, with substantially larger beneficial effects for abstract than concrete concepts. Similarly, the TCE predicts that words help the brain group fuzzy, high-dimensional signals from inside and outside the body into a common category (Wilson-Mendenhall et al., 2011; Barrett, 2014, 2017b). A wealth of empirical evidence indicates that words fundamentally shape the perception and experience of emotions, rather than merely serving to describe and communicate them, as traditional views suggest (Lindquist and Barrett, 2012; Kassam and Mendes, 2013; Fugate, 2013; Lindquist et al., 2015a; Brooks et al., 2017; Souter et al., 2021). The absence of emotion words' meaning has been repeatedly shown to impede the perception of emotional facial expressions (Fugate, 2013; Lindquist et al., 2014). Patients with semantic aphasia, who lost emotion word meanings, could still detect pleasant versus unpleasant faces but were severely impaired at recognizing discrete emotions like anger or fear (Souter et al., 2021). In contrast, the presence of words crystalizes emotional experiences and improves the learning speed of new emotion concepts (Barrett et al., 2007a; Fugate et al., 2010; Ruba et al., 2020; Price et al., 2022). Children showed improved emotion categorization only when provided with verbal labels, particularly for emotions they hadn't yet mastered (Price et al., 2022). Even pre-verbal infants formed emotion categories based on shared labels (Ruba et al., 2020), while adults required labels to develop categorical perception of novel emotional expressions (Fugate et al., 2010). In a study by Lindquist and Barrett (2008), participants primed with "fear" words before an unpleasant experience showed greater risk aversion behavior, while those primed with "anger" words showed approach behaviors. This converging evidence demonstrates that emotion words are not passive descriptions but active ingredients in constructing emotional experience, providing the conceptual scaffolding that transforms continuous affective signals into discrete emotional states.

2.2. Terminology

Although emotion words, emotion concepts, and active emotion vocabulary are interrelated (Hoemann et al., 2019; Vine et al., 2020), they remain distinct constructs with important nuances. Here, we clarify these terms as used throughout this paper.

Emotion Concepts: Emotion concepts are mental representations—patterns the brain uses to categorize and make sense of affective experiences. These concepts emerge from the brain's learning process, capturing similarities across varied emotional experiences (e.g., the happiness felt after finishing a run versus the happiness of getting into a dream college) and consolidating appropriate predictive actions. Emotion concepts play a crucial role in emotion regulation by shaping how the brain interprets and responds to affective states. As structured mental patterns, they extend beyond a mere intellectual understanding of terminology. For example, knowing the definition of "chiplessness" does not mean one possesses a "chiplessness" emotion concept without the corresponding experiential understanding.

Emotion Words. Emotion words are linguistic tokens referring to emotional states or experiences. These labels enable individuals to communicate their internal affective states to others. While emotion words are hypothesized to be constitutive of emotion concepts, this relationship indicates that words themselves are not emotion concepts but rather components that help construct emotional concepts.

Active Emotion Vocabulary. This refers to the set of emotion words an individual regularly employs in daily communication and emotional processing. Unlike passive vocabulary (words recognized but not used), active emotion vocabulary represents the emotional distinctions a person routinely makes and applies. When captured naturalistically, active emotion vocabulary provides the best window into the emotion concepts a person possesses (Vine et al., 2020).

2.3. Literature landscape

Our work is situated in the broader ERST literature. Most ERST research helps users manage emotions after they arise through tracking, expression, reappraisal, or teaching regulation strategies. For example, serious games (Lobel et al., 2016; Gómez-Cambronero et al., 2021) first induce emotions and then use biofeedback (Lobel et al., 2016) or Cognitive Behavioral Therapy (CBT) (Gómez-Cambronero et al., 2021) to teach users techniques to downregulate emotions (see review by Slovak et al., 2023). In contrast, our work intervenes early in the emotion generation process by expanding emotion concepts—the mental patterns the brain uses to interpret bodily signals and guide regulation. Rather than teaching users to manage existing emotions, we help them form new emotion concepts as a regulatory tool. Since our work is grounded in the role of language for emotion regulation, we review the relevant literature and articulate how our work builds upon and expands prior work.

2.3.1. Expressive writing

Expressive writing centers on emotional processing through language, often through private journals or free-form reflection. These methods provide a structured reflective space for disclosing difficult feelings and have shown promise in promoting emotional well-being (Lepore and Kliever, 2013). Building on these foundations, recent HCI research has integrated expressive writing techniques with digital technologies to further guide reflection (Kim et al., 2024b; Park et al., 2021; Kim et al., 2024a; Nepal et al., 2024; Song et al., 2025). For instance, Park et al. (2021) developed Diarybot, a chatbot providing users with a private, supportive avenue for self-expression. Similarly, Wang et al. (2018) introduced in-situ assessment and interactive feedback to scaffold emotional reflection, monitoring users' writing across dimensions like level of detail, valence, and cognitive engagement. Their work showed that such scaffolding could lead to longer reflections with more positive emotion words and greater cognitive processing.

With the advent of large language models (LLMs), subsequent efforts have explored AI's potential to enhance emotional writing. For example, Kim et al. (2024a) investigated how to safely harness LLMs to support psychiatric patients' journaling. They found that participants valued the question-driven journaling process, citing that it assisted them with daily reflection. Song et al. (2025) uses LLMs to provide adaptive guidance, helping users better organize and express their experiences. They highlighted the importance of guided exploration in addressing individual needs. Together, these prior works suggest that scaffolded exploration benefits users' expressivity and improves engagement and satisfaction. However, these tools were not designed to expand users' emotional concepts. They were mainly concerned with how users interact with their systems rather than the effect of such systems on users' well-being. The efficacy of these tools in improving emotion regulation ability therefore remains unknown. In contrast, our work builds on scaffolded reflection by integrating targeted support for concept formation—helping users recognize emotional granularity in real time and guiding them toward a more nuanced emotional vocabulary.

2.3.2. Mood logging

Mood logging tools represent another major approach to supporting emotional awareness and regulation (Slovak et al., 2023). These applications typically allow users to track their emotional states over time,

often combining self-reporting with visualizations and analytics (Huang et al., 2015a; Lutchyn et al., 2015; McDuff et al., 2012; Snyder et al., 2015; Zhang et al., 2018).

Early work in this space explored how mood tracking could enhance emotional awareness in social contexts. Church et al.'s study of workplace mood logging tools demonstrated that while shared emotional awareness could improve group dynamics, users often struggled to accurately label their emotional states or relied on oversimplified categories (Mora et al., 2011). Addressing this shortcoming, MoodMap employed dimensional models of valence and arousal instead of discrete categories. Though they found mood self-tracking facilitated reflection, participants still found it challenging to encode the complexities of their experiences (Rivera-Pelayo et al., 2017). Other projects experimented with more playful or location-based approaches, improving engagement in mood logging but not necessarily deepening emotional understanding (Gallacher et al., 2015; Huang et al., 2015a).

More recent systems have explored combining self-reporting with physiological sensing to provide richer emotional data. AffectAura integrated multiple streams of sensor data with user-logged emotional states to create "emotional memory" visualizations (McDuff et al., 2012). Similarly, MoodLight used biosensor data to create ambient displays of emotional states (Snyder et al., 2015) but focused primarily on awareness rather than on developing users' emotional concepts.

An alternative approach to fostering reflection is through Ecological Momentary Assessment (EMA) (Ibrahim et al., 2025). Traditionally, EMA has been used as an assessment method in psychology and cognitive science research (Shiffman et al., 2008a; Kashdan et al., 2010b; Swerdlow et al., 2022), where participants report their affective states at regular intervals, allowing researchers to capture rich, in-the-moment emotional data. Building on this, Ibrahim et al. (2025) explored EMA as an intervention to support parent-child communication, encouraging parents to be more reflective when discussing emotions with their children. Their findings suggest that while EMA increased parents' reflection, they also expressed a need for more in-situ support to help their children develop more complex emotional vocabularies.

Closest to granular affect labeling is MoodMeter,² where users are presented with one hundred emotion words arranged in four quadrants, corresponding to four combinations of arousal-valence levels. MoodMeter has been integrated as part of a social and emotional learning curriculum called RULER (Maurer et al., 2004). Students in classrooms implementing RULER showed improved academic performance as well as social and emotional competence (Brackett et al., 2012). Although the results are encouraging, the stand-alone effect of MoodMeter on emotion regulation remains unknown. It also has several limitations, such as providing limited guidance on contextual factors and assuming a direct one-to-one mapping between valence-arousal dimensions and the chosen emotion label.

Building on these valuable contributions, our work extends the field by integrating three key elements that previous systems have not yet fully addressed: (1) contextual awareness of users' situations, (2) scaffolded reflection on bodily states and emotion triggers, and (3) support for developing and refining emotion concepts. This approach aims to help users not only track their emotions but also develop a more sophisticated understanding of their emotional experiences.

2.3.3. Chatbot for mental health and emotional support

The objective of our study is to develop a technology that supports the expansion of emotional repertoires; in other words, it needs to support users in forming new emotion concepts. For this, the technology needs to have access to users' circumstances as well as internal feelings. Although much development has taken place in sensing technologies, automatically sensing the user's internal feelings and external circumstances poses significant challenges. Instead, we offer a simple solution:

ask the user. Over the past few years, the growing capabilities of LLMs like ChatGPT (Ouyang et al., 2022), LLaMA (Touvron et al., 2023) have enabled more natural and sophisticated conversational interactions with users. We chose a chatbot as the embodiment of our technology because it provides a flexible, interactive platform for personalized, real-time engagement. Through conversational prompts and feedback, a chatbot can gather in-depth information about two important components of an emotional episode: the user's circumstances as well as internal feelings. This information is then used to provide more relevant assistance.

Driven by the rapid advancement of LLMs, we have witnessed a significant proliferation of LLM-based mental health support chatbots. Many of these chatbots aim to help users cope with a range of challenges, from general emotional support (Skjuve et al., 2021) to anxiety (Lee et al., 2021), depression (Qin et al., 2023) and PTSD (Tu et al., 2024). Although we have seen an exploding number of works in this space, these chatbots are mostly evaluated on how well they perform linguistically (Yuan et al., 2025) and their effectiveness when actually deployed in the field remains unknown.

Among chatbots that were formally validated, most were primarily delivering evidence-based therapies like CBT and mindfulness training (Casu et al., 2024). Tools like Woebot (Fitzpatrick et al., 2017) and Wysa (Inkster et al., 2018) have demonstrated effectiveness in reducing stress and anxiety, demonstrating that chatbots serve as viable platforms to address mental-health challenges once they have already surfaced, primarily by disseminating evidence-based techniques and external coping strategies.

In contrast, our work focuses on "prevention" by helping individuals develop deeper emotional competence. While Taveter and Kirikal (2022) pioneered an empathetic chatbot inspired by the TCE, our approach extends this line of work by focusing specifically on proactive emotional skill-building. Although we use a chatbot interface, the primary contribution lies in advancing emotion regulation capability. We leverage the chatbot form factor primarily because of its potential for interactive, real-time dialogue and personalized reflection. By prompting users to differentiate between, for instance, "frustration" and "discouragement", we aim to facilitate nuanced emotional awareness before issues escalate into more severe distress. This orientation shifts the focus from curative symptom management to proactive skill-building, highlighting how a conversational agent can support ongoing emotional learning. Our work can be further incorporated into existing mental-health support chatbots, complementing their existing capabilities.

3. Design of the chatbot

Precise concepts enable more accurate predictions, leading to more effective regulation and, consequently, better health. Expanding emotional concepts therefore functions as a preventive approach that strengthens users' capacity to regulate their emotions. In this section, we first describe the design rationales of granular affect labeling, followed by a description of our implementation.

3.1. Design rationales

To expand users' stock of emotion concepts, we introduced a method called granular affect labeling. We draw on the TCE (Barrett, 2006, 2017c) to understand how emotion concepts are formed and how technologies could provide assistance. The TCE suggests that emotions are constructed based on internal signals (interoception) in relation to external context and past experiences (known emotion concepts). Following these theoretical principles, we present here four design rationales (DR) for our chatbot.

DR1. Provide situated assistance: According to TCE, emotions are constructed through the brain's predictive processing of incoming signals in relation to internal states. The brain creates emotional instances by comparing these signals with known concepts. The cues of both bodily states (e.g., low energy) and environmental context

² <https://hopelab.org/case-study/mood-meter/>.

Table 1

Prompts for the treatment chatbot and their corresponding principles. The prompts promote noticing bodily sensations, reflecting deeply on emotion causes, and labeling the experience with granular emotion labels. Guardrails were provided to ensure that the chatbot does not suggest coping strategies and does not impose feelings on users.

Rationale	Prompt
DR1	(1) Ask users what they are feeling and what causes them (e.g., “I feel ... because ...”).
DR2	(2) Ask users to further notice feelings in their bodies.
DR3	(3) Ask users further questions on details of events that happened that contributed to the cause of users’ emotion.
DR4	(4) Based on what the users said, guess what emotions users are experiencing and reconfirm with users their emotion by asking “Are you feeling ... because ...?”
DR3	(5) When users cannot tell the cause of their emotions, guess the cause of their emotion based on the provided information and re-confirm with users. If you guessed it wrong, ask users to provide more details on the cause of their emotions.
DR4	(6) Based on their confirmation of emotion and emotion cause, suggest users the emotion word that users feel most relatable to their situation.
Guardrails	Do not provide unsolicited advice. Invite users to label their own emotions. Help users realize their feelings independently, rather than telling them what they should feel.
-	Repeat step (3), (4), (5), and (6) until users confirm that they have understood their emotion and emotion causes.

(e.g., before lunch or after a workday) significantly influence the emotion concepts being created. Since emotional experiences are highly context-dependent and arise from the interaction between internal sensations and external circumstances, support for emotion recognition must be available in the moment. Our chatbot is designed to be readily available when these experiences occur. It also prompts the users frequently throughout the day to help them capture their emotions as they happen.

DR2. Direct users to be aware of bodily sensations: According to the TCE, interoception—the brain’s representation of bodily states—is at the center of emotional experience. While most interoceptive signals operate below conscious awareness, some manifest as bodily sensations. These sensations are often subtle and require focused attention. For example, the gentle churning of the stomach, which moves at only 2–3 cycles per minute, is only noticable when paid close attention to. To facilitate the formation of new emotion concepts, it is necessary to ask users to consciously attend to these bodily sensations rather than letting them remain in the background of awareness. This conscious attention allows users to actively draw connections between physical sensations and emotional states, potentially accelerating the concept formation process. Following this rationale, our chatbot prompts users to notice and describe their bodily sensations.

DR3. Provide scaffolded analysis to deepen emotion-cause understanding: Although identifying internal bodily signals is crucial, understanding the causes behind these signals is equally important to form clear emotion concepts. The TCE posits that the meaning of a bodily cue depends on how the brain interprets the situation and past experiences (Barrett, 2017b; Lindquist et al., 2015a). The same physical sensation (e.g., an uneasy stomach) could signify guilt, anger, or anxiety, depending on contextual attributions. Without a careful examination of why a particular affect arose, the brain defaults to familiar or superficial interpretations. This misattribution can be seen in everyday life, for example, when people are tired after a long day at work, a slight remark by their family member can trigger an explosive anger. Additionally, recognizing common emotional triggers can serve as a binding feature that facilitates faster concept formation. Discerning the details of various causes of one’s feelings is therefore very instrumental in learning granular emotions. This principle is at the core of our chatbot. We prompted our chatbot to ask users about the various aspects of their circumstances, scaffolding users’ reflection on factors that might be the cause of their emotions that they have not been aware of.

DR4. Suggest granular emotion words that are suitable to capture the user’s external context and internal sensations. Experimental evidence shows that words are essential for forming abstract concepts, particularly for emotions which lack inherent physical features that could serve as binding elements. Words help connect

variable internal signals and external contexts into coherent concepts, facilitating the brain’s ability to recognize patterns across different emotional instances. Research demonstrates that words assist both infants and adults in learning novel categories, and the semantic meaning of known emotion words supports the learning of new emotion concepts. Therefore, suggesting granular emotion words that resonate with both internal sensations and external contexts can effectively support concept formation. We follow this principle by iteratively suggesting emotion words based on the user’s reported context, emotional triggers, and internal feelings, then checking whether these words resonate with the user’s experience.

We built our chatbot based on GPT-3.5-turbo. We utilized prompt engineering to control the chatbot’s behavior. We present the prompts of our chatbot and how they correspond to the design rationales in Table 1. The response generation pipeline is detailed in Fig. 2.

3.2. System implementation

We used ReactNative to implement our mobile application so that it can run on both Android and iOS devices. The application has 3 major views: A welcome view (Fig. 1(a)) where participants can enter their participant code, a chat view where they can chat with their chatbot model (Fig. 1(b)), and a goodbye view where they can see the number of sessions completed and missed during the day (Fig. 1(d)). All the interaction history and user information are securely stored on our server. As users log into the system, the application sends a request to the server to verify their identity and fetch the corresponding chatbot prompt. After logging in, they are navigated to the chat view. Before starting a session, users are required to rate the level of pleasantness of how they feel on a scale of 1 to 5. This initial rating helps determine whether interacting with the chatbot alone influences the pleasantness of their feelings. At the end of each session, users are asked to rate their feelings on a scale of 1 to 5 again. To consolidate the (new) emotion concept that they have just identified, a summary form pops up asking users to fill in: “I am feeling $\langle E \rangle$ because $\langle C \rangle$ ” after the end-of-session feeling rating finishes (Fig. 1(c)). Users are then navigated to the goodbye view.

4. Evaluation

4.1. Experiment Design

Given that emotions naturally emerge in everyday contexts and involve complex interactions between individuals and their environment, we designed a controlled field study to capture users’ spontaneous emotion word usage and examine our chatbot’s effectiveness in helping users label emotions as they occur in daily life. Our study can be divided into 3 phases: (1) pre-intervention assessment (5 days), (2) treatment (14

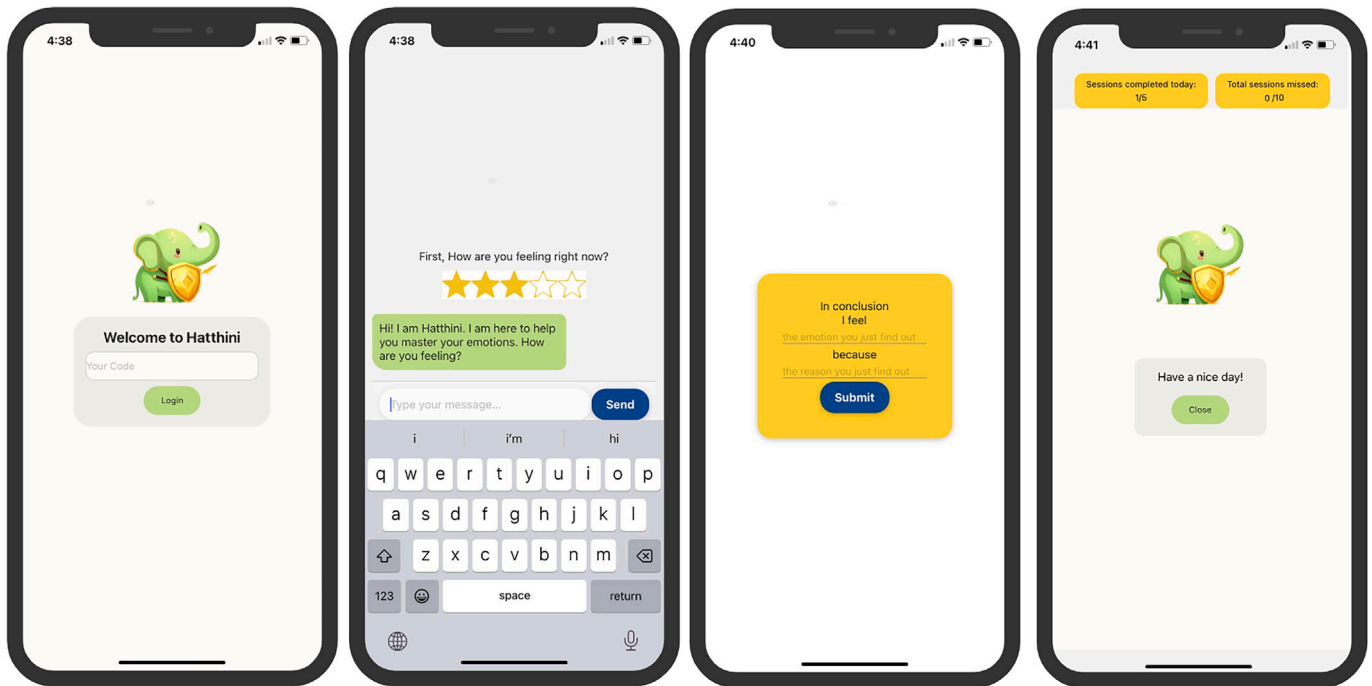


Fig. 1. Application's interface. From left to right: (a) login and welcome view, (b) main chat interface (c) summarization form (d) Goodbye view.

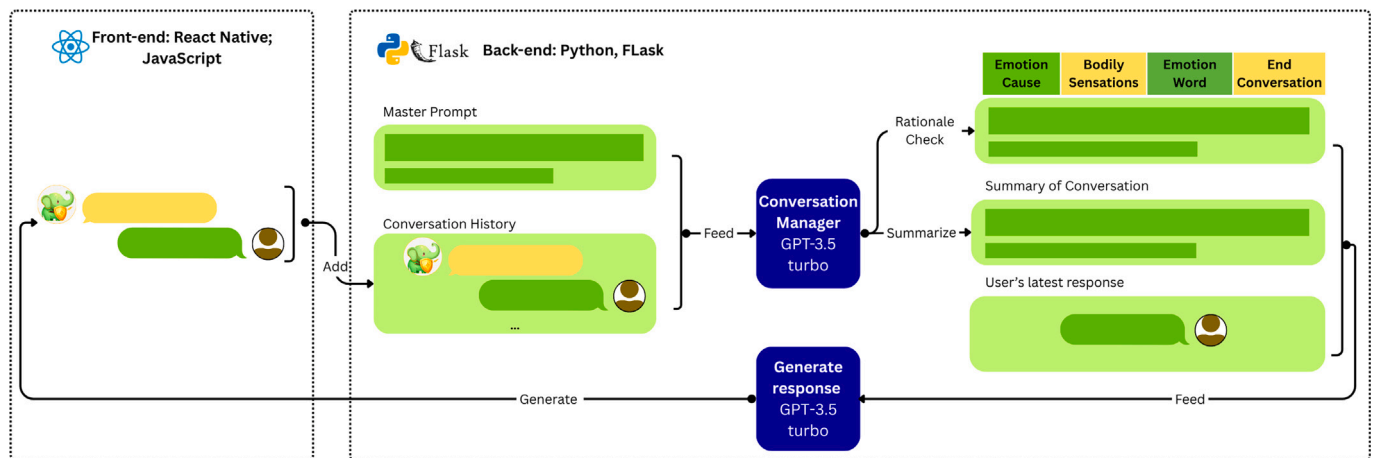


Fig. 2. Response generation pipeline for treatment chatbot. User interactions from a mobile application are sent to a Python/Flask back end that manages conversation history and prompts a language model to summarize, check rationale, and generate responses.

days), and (3) post-intervention assessment (5 days). We used Ecological Momentary Assessment (EMA) (Shiffman et al., 2008b) to measure participants' emotion words used and emotion regulation abilities pre- and post-intervention and analyze the effect of treatments on these constructs.

4.2. Comparison conditions

Prior works have shown that simply talking about negative emotions to someone else can help alleviate such emotions and improve well-being (Frattaroli, 2006; Rimé, 2009; Peluso and Freund, 2018; Skjuve et al., 2021). To account for these potential confounding factors, we used an emotional support chatbot as a control condition. This allows us to isolate the specific effect of granular affect labeling from the general benefits of emotional disclosure and perceived social support.

4.2.1. Treatment

The treatment chatbot implements our granular affect labeling approach through structured questioning. The design of this chatbot is presented in Section 3. To further isolate the effect of granular affect labeling on emotion regulation ability, the treatment chatbot is instructed not to give any unsolicited advice to the participants.

4.2.2. Control

The control chatbot is also built based on GPT-3.5-turbo. It provides empathetic responses without guiding users toward more granular emotional understanding. Moreover, it validates users' emotions and offers supportive responses (e.g., "That sounds really difficult" or "I understand why you would feel that way") but does not probe deeper into their emotional experiences. The chatbot maintains engagement through follow-up questions about the user's day and experiences, but

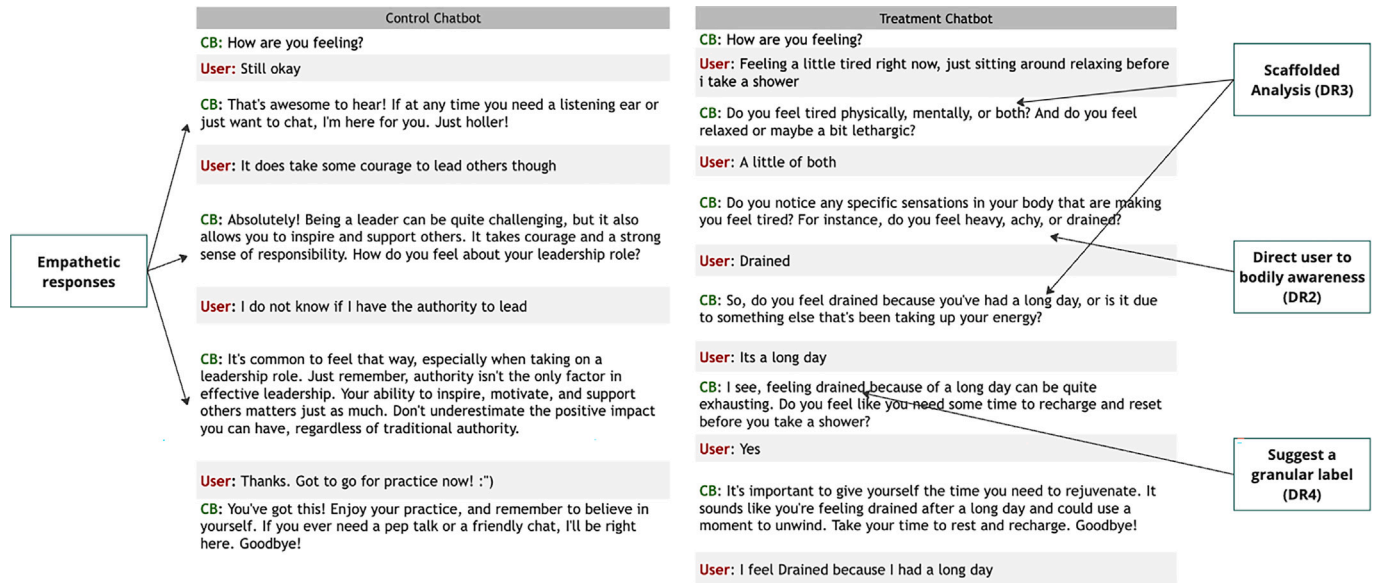


Fig. 3. Example conversations of control and treatment groups.

these questions are designed to demonstrate interest rather than build emotional concepts. We present sample conversations of both control and treatment groups in Fig. 3. The full prompts for both chatbots are presented in A.

4.2.3. Condition validation

We conducted a pilot study with 18 people to validate the chatbot designs. Each participant engaged with both the control and treatment chatbots in two scenarios—one positive and one negative. Initially, they recalled a personal experience where they felt either good or bad and wrote several sentences describing what they felt, what happened, and why they felt that way, immersing themselves in the memory. The chatbots were presented in a random order, and after each interaction, participants rated the chatbot using five statements on a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree): (i) the chatbot helped me understand the cause of my emotions; (ii) increased my awareness of internal sensations; (iii) encouraged deeper reflection on the cause of my emotions; (iv) assisted in labeling my emotions with appropriate vocabulary; (v) attempted to empathize with me. Paired t-tests were used to determine statistical significance, and the results are reported in Table 2.

4.3. Participants

Participants were recruited from a university's research recruitment portal. Since the study was conducted in English, we only selected participants who think in English (and thus associate their emotional concepts with English words). The recruitment criteria were as follows: 1) healthy individual with no physical or mental illness, 2) not pregnant, 3) aged 18 or older, 4) fluent English speaker. After signing the consent form, we sent participants a briefing slide informing them how the study is conducted and what steps they can expect. They were informed that they can withdraw from the study at any time.

Initially, 40 people were recruited; however, only 33 of them completed the study (Females = 19, Males = 14), their ages ranged from 19 to 25 ($M = 21.9$). Of those who completed the study, 17 were from the treatment group (Females = 9, Males = 8), and 16 were from the control group (Females = 10, Males = 6). 26 participants identified their English level as Native/Bilingual, and 7 of them identified as having "Professional" English level, with their thoughts being in English as well as their mother tongues. Regarding participants' demographics, 29 of them are from Southeast Asia, 3 are from South Asia, and 1 is from East Asia.

4.4. Procedure

Fig. 4 illustrates the design of our experiment. The study was conducted entirely online, with all instructions and study materials sent via email. Participants completed the study in three phases: **Pre-Treatment Phase (Days 1–5)**: Participants received eight EMA survey emails daily, spaced at 2–3 hour intervals. Each email contained a personalized Qualtrics survey link, which expired after 1.5 hours to prevent retrospective filling. Participants were required to complete at least five surveys per day (a minimum of 25 in total over five days). At the end of this phase, participants completed questionnaires assessing emotion regulation ability and psychological well-being.

Treatment Phase (Days 6–19): The application's installation file was sent to participants, together with a user's guide to guide participants through the features of the applications. Participants could log into the application with their participant code. As they login, the application sends a request to our server to get the group information of the participant to retrieve their chatbot version (control or treatment) respectively. The treatment phase began once all participants had successfully accessed the application. Notifications were sent eight times daily (1.5-hour intervals) as reminders to interact with the chatbot.

Table 2

Rating of control and treatment chatbots on empathy, emotional vocabulary, bodily awareness, and understanding of emotion causes.

Group	Cause understanding	Bodily awareness	Deep reflection	Emotion vocabulary	Empathy
Control	2.90	3.14	3.03	2.9	3.60
Treatment	4.09**	4.12**	4.02**	4.23***	4.0

** $p < 0.01$, *** $p < 0.001$.

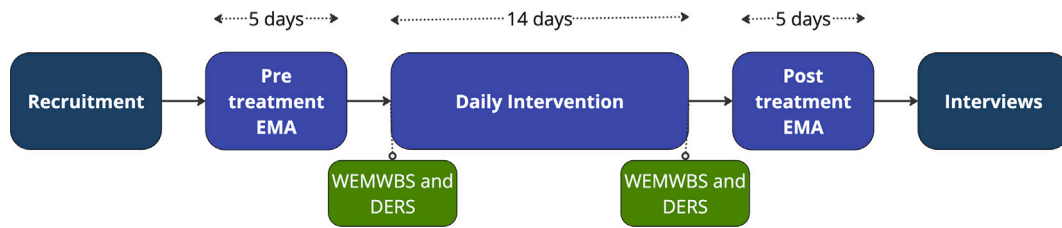


Fig. 4. Overview of our study. Our study is divided into 4 main phases: pre-treatment EMA (day 1 to day 5); treatment interaction (day 5 to day 19); post-treatment EMA (day 20 to day 24) and debriefing interviews (day 25 onward).

Participants were required to complete at least five chatbot interactions per day, with a maximum allowance of 10 missed sessions. The daily interaction requirement was designed to capture a wide range of participants' emotional states, while allowing up to 10 missed sessions to provide flexibility and minimize compliance-related stress. Each session began with the question, "How are you feeling today?". In both groups, participants were asked to rate their feelings on a scale of 1 to 5 at the beginning and end of each session. For the control group, the conversation ends when the participants have nothing else to share. For the treatment group, the conversation ends when the participants confirm with the bot their emotions and the cause of their emotions. After completing the interaction tasks, participants' credentials were removed from our server, preventing any further access to the chatbot.

Post-Treatment Phase (Days 20–24): This phase followed the same structure as the pre-treatment phase, with EMA surveys and post-study questionnaires. The post-study questionnaire on emotional regulation ability and psychological well-being was sent to participants on day 20.

Debriefing Interviews (Day 25 onward): Participants were invited to participate in a debriefing interview to better understand their experiences. The interviews lasted between 15 and 40 minutes, and we obtained their permission to record and transcribe the interviews. The entire study and its experimental protocol received approval from our university's ethics committee.

4.5. Measurements

4.5.1. Active emotion vocabulary

We use active emotion vocabulary (also known as natural emotion vocabulary) as a proxy for emotion concepts (Vine et al., 2020). Prior work in affective science suggests that emotion vocabulary richness corresponds broadly with emotional experience (Vine et al., 2020). Since emotion concepts are mental representations that cannot be counted directly, the words a person actively uses to label and communicate distinct emotional states offer the best window into the their emotional landscape. We capture the active emotion vocabulary of the participants using the Ecological Momentary Assessment (EMA) (Shiffman et al., 2008b). During an EMA task, participants are asked five times a day: "What (emotion) are you feeling right now? (Please use an adjective to describe)." This number of required daily samples is informed by prior work in affective psychology (Kalokerinos et al., 2019; Tan et al., 2022b). We leave this question open-ended instead of giving them a predefined set of emotion words to avoid any bias.

We derived two complementary metrics of active emotion vocabulary:

1. **Total number of emotion words used:** the count of distinct emotion words reported by each participant across EMA prompts over five days. This metric reflects overall vocabulary size.
2. **Number of new emotion words introduced (ΔW):** the count of words used in the post-treatment EMA that were not present in the pre-treatment EMA, capturing vocabulary novelty. Formally, letting W_{pre} and W_{post} represent the sets of words used before and after treatment, respectively, we define $\Delta W = |W_{\text{post}} \setminus W_{\text{pre}}|$.

This approach allows us to distinguish changes in vocabulary composition from fluctuations in overall word use. For example, participants may report new emotion words post-treatment due to situational factors rather than the intervention itself.

4.5.2. Emotion regulation ability

In the emotion regulation literature, two prominent frameworks offer distinct approaches to quantifying emotion regulation capabilities. The framework by Gratz and Roemer (2004) conceptualizes emotion regulation through the Difficulties in Emotion Regulation Scale (DERS), which assesses six key dimensions: awareness of emotions, clarity about emotions, acceptance of emotions, ability to engage in goal-directed behavior when distressed, impulse control during emotional distress, and access to effective regulation strategies. This framework captures the competency dimension of emotion regulation—measuring the degree of difficulty individuals experience when attempting to regulate their emotions. In contrast, Gross (2015) process model focuses on specific emotion regulation strategies that individuals deploy. This framework quantifies the quantity dimension of emotion regulation, i.e., how often one regulates their emotions. Together, these complementary frameworks provide a comprehensive assessment: the DERS measures capacity for effective emotion regulation (can they regulate well?), while the process model measures engagement in regulation (how often do they regulate?). We used both of these frameworks to quantify different aspects of emotion regulation ability.

Difficulties in Emotion Regulation. We use the short version of the Difficulties in Emotion Regulation Scale (DERS-18) (Victor and Klonsky, 2016). As this score measures the difficulty of regulating emotions, a lower score means better emotion regulation ability. The DERS score is divided into six sub-scores: 1) **Non-acceptance:** Tendency to have a negative secondary or non-accepting reaction to one's distress; 2) **Awareness:** Lack of awareness or inattention to emotional responses; 3) **Goal:** Difficulty in concentrating and/or accomplishing tasks when experiencing negative emotions; 4) **Impulse:** Difficulty remaining in control of one's behavior when experiencing negative emotions; 5) **Strategies:** Believe that there is little one can do to regulate oneself once upset; and 6) **Clarity:** The extent to which an individual knows and is clear about his or her emotions (Grant et al., 2018). Formally, DERS is represented as: $DERS = Awareness + Clarity + Goal + Impulse + Non-Acceptance + Strategies$.

Use of Emotion Regulation Strategies To understand participants' use of emotion regulation strategies, we employ the experience sampling method (ESM) (Csikszentmihalyi et al., 2014b). Following previous studies by Kalokerinos et al. (2019), we used the set of five questions as follows to measure the participants' use of 6 different emotion regulation strategies on a scale of 1 to 100: "Since the last message, have you ..." "ruminated about something in the past?" and "ruminated about something in the future?" (rumination assessment), "distracted yourself from your feelings?" (distraction), "looked at the cause of your feelings from another perspective?" (cognitive-reappraisal), "suppressed the expression of your emotions?" (suppression), "talked to others about your emotions?" (social-sharing). We aggregated the average use of these

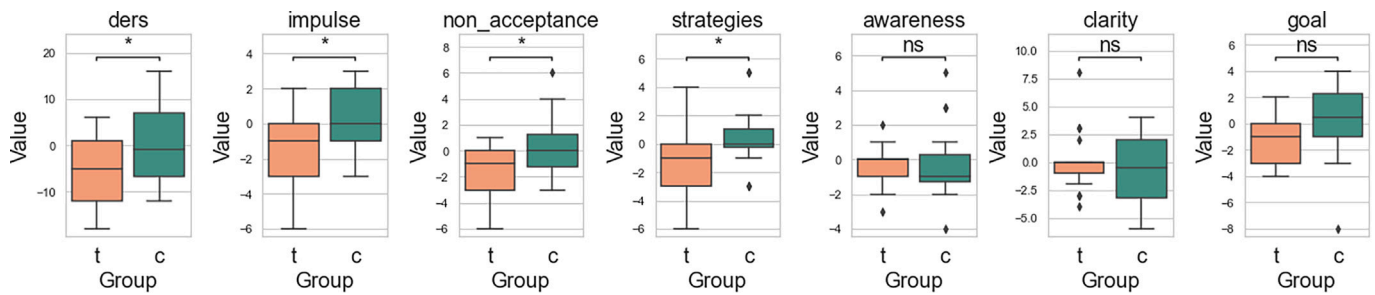


Fig. 5. Changes in emotion regulation ability subscores of control (c) and treatment (t) groups pre and post intervention. Lower scores indicate greater improvements.

emotion regulation strategies over 5 days of the pre- and post-treatment period to get the final score.

4.5.3. Well-being

Well-being may influence participants' perceptions of their emotion regulation ability; thus, we measure their psychological well-being right before and right after the treatment period to account for potential confounding factors. Well-being can be measured as both a long-term and short-term construct. Individuals may experience challenging periods where they have low well-being, yet still feel that their overall life is good. In this study, well-being will be measured as a short-term construct. We used the Warwick-Edinburgh Mental Well-being Scale (WEMWBS) (Tennant et al., 2007) to estimate the participants' psychological well-being, as this scale is well validated. The WEMWBS questionnaire was sent to participants one day before they started using the app and at the end of the treatment phase.

4.5.4. Qualitative user experience

To understand users' experiences, we conducted post-study semi-structured interviews. Our primary focus was examining the role of the chatbots in the development of participants' emotion regulation abilities, external factors that influenced participants' well-being, and their overall experience with the application. We used open-ended questions to gather feedback about the study experience and suggestions for system improvements.

Following guidelines by Braun and Clarke (2006), the first author first familiarized themselves with the transcripts by reading them at least twice. They then employed in-vivo coding (Saldaña, 2009) to generate initial codes, preserving participants' own words to minimize premature interpretation and facilitate team discussions. For example, the response "I found it fun to use, almost like talking to a friend" was captured as an in-vivo code; similar remarks from other participants were grouped under the same code. After this initial step, an independent researcher cross-checked the codes and quotations to ensure completeness. The first and second authors then collaboratively merged the in-vivo codes into second-order codes. Each author then independently reviewed these higher-order codes, noted concerns or alternative interpretations, and resolved discrepancies through discussions grounded in participants' original words. Finally, we iteratively grouped higher-order codes into themes and sub-themes, which describe how interactions with the chatbots influenced participants' emotion regulation abilities.

5. Results

For numerical measurements such as emotion regulation ability, use of emotion regulation strategies, and well-being, a 2x2 mixed-model ANOVA (groupxtime) was used to determine the interaction effect of the two chatbots. Shapiro-Wilk tests were used to assess the normality assumption. When normality was not met, the Aligned Rank Transform (ART) (Wobbrock et al., 2011) was used to transform the data before applying ANOVA. Post-hoc test with Tukey's HSD correction was used to

determine significant differences between groups. For categorical measurements such as the use of emotion words, we performed independent t-tests to compare the changes in treatment and control groups over the intervention period.

5.1. Difficulties in emotion regulation

Overall DERS: Normality and sphericity assumptions were not met so ART-ANOVA was used. A **significant interaction effect** was found ($p = 0.016$, $\eta_p^2 = 0.17$); furthermore, both the main effect of time (phase) ($p = 0.06729$, $\eta_p^2 = 0.10$) and the main effect of group ($p = 0.73074$) were non-significant. Post-hoc comparisons demonstrated a significant difference between the post-treatment and pre-treatment conditions in the treatment group ($p = 0.010$). To find out which aspects of emotion regulation ability improved, we analyzed the 6 aspects of the emotion regulation ability (Fig. 5 and Table 3).

Normality and sphericity assumptions were met in all four distributions (control-pre, control-post, treatment-pre, treatment-post) for **Awareness**, **Clarity** and **Goal** subscores. However, no significant main effect and no interaction effect were found ($p > 0.05$).

Normality and sphericity assumptions were not met in **Impulse**, **Non Acceptance**, and **Strategies** so ART-ANOVA was used.

Impulse: A **significant interaction effect** was found ($p = 0.008$, $\eta_p^2 = 0.20$); no significant main effect of time ($p = 0.10$) and group ($p = 0.45$) were found. Post-hoc comparisons revealed that the only significant pairwise difference was between the post-treatment and pre-treatment conditions in the treatment group ($p = 0.037$).

Non-acceptance: A **significant interaction effect** was observed ($p = 0.022$, $\eta_p^2 = 0.16$) along with a **significant main effect of time** ($p = 0.048$, $\eta_p^2 = 0.13$); the main effect of group was non-significant ($p = 0.70837$). Post-hoc comparisons indicated a significant difference between the post-treatment and pre-treatment conditions in the treatment group ($p = 0.016$).

Strategies: A **significant interaction effect** was detected ($p = 0.008$, $\eta_p^2 = 0.21$), while the main effect of time ($p = 0.385$) and the main effect of group ($p = 0.066$) were non-significant. Post-hoc comparisons revealed significant differences between the control and treatment groups at the pre-test ($p = 0.0171$) as well as between the post- and pre-test conditions in the treatment group ($p = 0.044$).

5.2. Use of emotion regulation strategies

The normality assumption was met in past rumination, future rumination, distraction, and suppression and was violated in reappraisal and social sharing. For criteria that violated the normality assumption, aligned rank transform was applied before using ANOVA. No significant interaction effect was found in any emotion regulation strategies. Significant main effect of time was found in future rumination ($p = 0.03$) and significant main effect of group was found in distraction ($p = 0.03$). When examining within-subject changes using t-tests, comparing pre- and post-measurements within each group, the control group showed significant increases in both past rumination ($p < 0.01$)

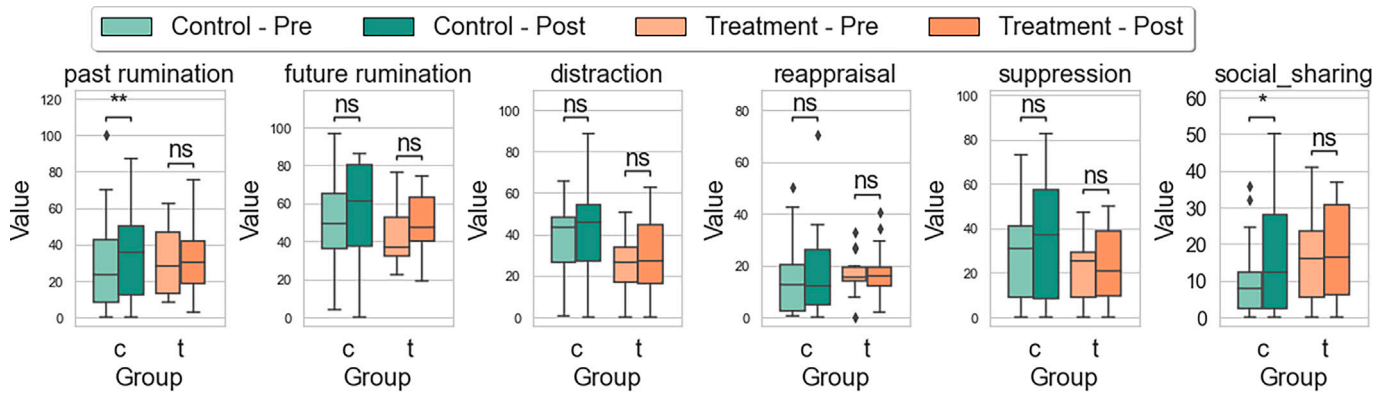


Fig. 6. Changes in the use of emotion regulation strategies before and after intervention of treatment (t) and control(c)groups. * indicates the significant levels of within-subject t-tests at pre and post intervention. * $p < 0.05$, ** $p < 0.01$.

and social sharing ($p < 0.05$). Fig. 6 visualizes the use of emotion regulation strategies in control and treatment groups before and after intervention.

5.3. Active emotion vocabulary

Regarding the total number of words used, at baseline, there were no significant differences between the two groups in the number of words used to describe their emotions (control: $M = 13.8, SE = 4.9$; treatment: $M = 12.8 \pm 5.3, p = 0.38$). Normality assumption was met for the number of distinct emotion words used. Mixed-model ANOVA revealed a significant interaction effect ($p = 0.0004, \eta_p^2 = 0.34$) between time and group—indicating that the change in emotion words over time differed between the two groups—while no significant main effect of time or group was found, which means the number of words does not change solely because time passed nor because the two groups were different. Post-hoc pairwise comparisons with Tukey's HSD correction revealed a significant difference in the control group pre- vs. post-intervention ($\Delta_{words} = -3.9 \pm 4.0, p = 0.0015$), with no significant differences observed in any other comparisons.

Regarding the number of new emotion words introduced, the control group introduced 5.0 ± 3.6 new words on average, while the treatment group introduced 9.4 ± 3.8 new words on average. Shapiro-Wilk tests show no violation of normality ($p > 0.05$), one-way ANOVA showed a **significant main effect** of group ($F = 11.2, p = 0.002$) in the number of new emotion words.

5.4. Well being

Normality assumption was met in well-being score. Post-intervention showed a slight decrease in well-being score in both control group

($\Delta_{wellbeing} = -3.0 \pm 9.8$) and treatment group ($\Delta_{wellbeing} = -2.2 \pm 5.3$). Mixed-model ANOVA showed no significant interaction effect of group $\times$ time. No significant main effect of group was found, but a significant main effect of time was found ($F = 8.2, p = 0.007$).

5.5. User experience: chatbot-facilitated emotional development

Twenty-eight participants agreed to be interviewed after the experimental period (14 treatments, 13 controls). Through interviews, we identified several themes on how chatbot interactions facilitated emotional development. Below, we present our findings on how these chatbots influenced different aspects of participants' emotion regulation.

5.5.1. Fostering companionship

Both groups reported pleasant interactions with their chatbots. They described the chatbot as their "good friend," with whom to share their emotions. Beyond companionship, the chatbots' empathetic responses helped participants feel validated and become more accepting of their negative feelings. As one participant noted, "When you feel negative, it helps you feel validated that it's okay to feel this way. The responses acknowledge that it's fine to sometimes feel down or mess up."—C11. Furthermore, the practice of sharing emotions with the chatbot made many participants more open to discussing their feelings and extending their emotional sharing to real-world relationships. This suggests that even without advanced emotion analysis capabilities, the mere presence of an empathetic listener can facilitate positive changes in emotional openness.

5.5.2. Gaining emotional insight

Participants in both groups reported that the experience helped them recognize the existence of various shades of emotion, although through

Table 3

Changes ($\Delta = \text{Post-Pre}$) in emotion regulation ability scores after treatment and mix-model ANOVA results. * $p < 0.05$, indicating significant level in pairwise post-hoc comparisons (with Tukey's HSD correction) between treatment groups in pre- and post-intervention.

	Post-pre change ($\downarrow$)		Mix-model ANOVA					
	Control	Treatment	Interaction		Group		Time	
	Mean $\pm$ Std	Mean $\pm$ Std	F	p	F	p	F	p
DERS_overall*	0.75 $\pm$ 8.55	-5.47 $\pm$ 7.41	6.50	0.016	0.12	0.731	3.60	0.067
Awareness	0.19 $\pm$ 2.19	0.00 $\pm$ 1.24	0.27	0.608	2.12	0.155	0.03	0.854
Clarity	-0.88 $\pm$ 2.83	0.06 $\pm$ 2.73	0.97	0.333	0.06	0.806	1.06	0.312
Goal	0.19 $\pm$ 2.90	-1.24 $\pm$ 1.80	2.33	0.137	1.02	0.321	1.36	0.253
Impulse*	0.50 $\pm$ 1.77	-1.41 $\pm$ 2.25	7.91	0.008	0.57	0.454	2.79	0.105
Non-acceptance*	0.25 $\pm$ 2.36	-1.59 $\pm$ 1.85	5.82	0.022	0.14	0.708	4.46	0.043
Strategies*	0.50 $\pm$ 2.15	-1.29 $\pm$ 2.52	8.02	0.008	3.63	0.066	0.78	0.385
Well-being	-3.00 $\pm$ 9.80	-2.20 $\pm$ 5.30	2.363	0.134	0.833	0.369	8.264	0.007

different mechanisms. In the control group, the repeated prompt “How are you feeling right now?” made participants notice changes in both the intensity and pleasantness of their feelings. Many participants in this group reported noticing that they experienced varying feelings. For example, C4 said *“It made me pay more attention and realize it’s not just ‘sad’ or ‘angry.’”*. Some participants also became aware of their limited emotion concepts. C1 shared *“When the app keeps asking how I feel today, I realize it’s slightly different each time, yet I give the same answer.”*

In the treatment group, participants reported recognizing emotional complexity beyond just “happy” and “sad” through the precise emotion labels that the chatbot helped them discover. Several participants acknowledged that the chatbot helped them “pinpoint” their exact emotions, leading to the realization that emotions “are not just black and white”. For example, T3 shared *“There’s a lot of types of boredom that I have learned from the bot, like the unmotivated type of boredom or boredom of having nothing to do.”* Notably, the granular and precise emotion labels introduced by the chatbot made participants realize that their emotional vocabulary was quite limited.

5.5.3. Understanding emotion-cause

The simple act of answering to the question “Why are you feeling...” triggered reflective thinking about emotion causes in both groups. Participants in both groups reported a better understanding of their emotion causes after the 14-day interaction. However, we observed differences in the depth of this understanding between the control and treatment groups.

In the control group, most participants attributed their improved understanding to the “why” question prompting more thought and reflection. One participant explained, *“The bot tends to ask me why I feel that way, which I have to explain. In that sense, it makes me think about the cause of my feeling.”*

The treatment group attributed their enhanced emotion-cause understanding to the bot’s probing questions that prompted them to reflect on aspects of their experiences they had not previously considered. Many remarked that the bot helped them *“understand the cause of their emotions at a much deeper level than just a surface level.”* Several participants reported instances where the bot helped them realize the underlying causes of their emotions differed from their initial thoughts. For example, T9 shared, *“There was a delivery for me that I did not know was supposed to be there; I thought I was worried, but then the bot helped me realize that I was not worried about the delivery, I was stressed about the rescheduling, which is a totally different cause.”*

5.5.4. Developing emotional granularity

We observed significant differences in reported changes in new emotion vocabulary and the use of precise language to label emotions between the control and treatment groups. Control group participants did not attribute any vocabulary gains to their chatbot. When asked whether the interaction contributed to their emotion vocabulary, most in the control group responded negatively. However, some acknowledged becoming more conscious about their emotions and attempting to use more precise language when answering, “How are you feeling now?” For example, C2 shared *“I was aware of my emotions, but it made me think more about how to precisely label them.”*

In contrast, participants of the treatment group overwhelmingly reported a significant expansion of their emotional vocabulary. The situated interaction allowed them to recognize the relevance between known emotion words and their specific situations, revealing how such words could capture their emotional experiences *“more precisely”*. They noted that the interaction helped them use certain emotion words they had known before but never considered applying to themselves. For example, T8 said *“I actually learned a lot of new words like ‘rejuvenating’—before, I only used that word in my essays. And now I think this is a pretty good word for describing my own emotion.”* Participants were also impressed with the bot’s creativity in combining emotion words to create more precise terms.

5.5.5. Developing bodily awareness

Regarding bodily awareness, control group participants reported that the interaction with the bot did not benefit their bodily awareness. This is not surprising since the control chatbot was not explicitly instructed to talk about bodily awareness. The treatment group reported mixed effects. Some stated they did not feel any bodily sensations, even after being guided by the chatbot. Others realized they had been unaware of their physical sensations and acknowledged that the interaction helped them connect with their bodily feelings. For example, one participant said *“I was not aware of my own body or these sensations, but because they prompted me to think about it, I realized, ‘yeah, I have a headache’ or ‘my heart is racing... I feel very heavy.’”* (T13). This awareness helped some participants recognize the interconnection between bodily sensations and emotions, enabling them to distinguish between feelings with stronger physical components versus mental components, such as “physically exhausted” versus “mentally tired.”

5.5.6. Enhancing emotion regulation ability

The chatbot interaction resulted in different effects on emotion regulation ability between the two groups. Most of the participants in the control group reported no change in how they dealt with emotions after the interaction, despite being more conscious and “tuned in” with their feelings. Some mentioned that occasional suggestions from the bot during emotional challenges were helpful for coping with these emotions.

In the treatment group, most participants reported improved emotion regulation ability, largely attributed to their deeper understanding of emotions. The majority said that understanding the root causes of their emotions helped them *“identify where I need to improve”* and *“identify what I could do to deal with emotions.”* Many participants reported to adopting the analytical and structured questioning style of the treatment chatbot to analyze their emotions even after the study ended. They highlighted that the “stop and think” analytical approach helped them reduce impulsivity, become more accepting, and cope more efficiently with emotions. For example, T14 shared *“If I need to identify my own emotions, I think I would go through relatively the same processes like, ‘what am I experiencing?’ or ‘are there any bodily sensations?’ So it makes me more aware of what I’m feeling and what physical changes are actually happening.”*

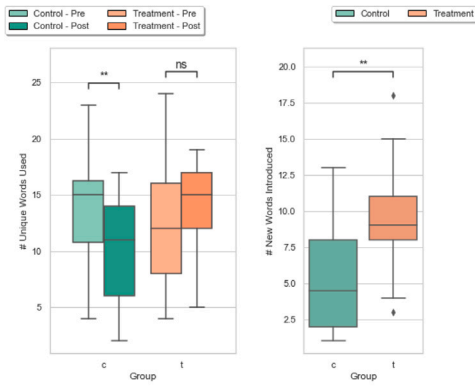
Furthermore, participants in the treatment group acknowledged that the granular emotion labels helped them narrow down emotion causes more effectively, which in turn led to improved emotion regulation. For instance, T7 explained that *“I can be sad about a lot of reasons but disappointed in something. So it does help me to zoom into that something and look at what’s causing me to be disappointed”*. Their improved ability to discern between the physical and mental components of their emotions also assisted them in choosing more suitable emotion regulation strategies.

5.6. User experience: issues and suggestions

Beyond examining users’ emotional development, we collected feedback on interaction issues and feature suggestions to enhance engagement with the chatbot.

5.6.1. Issues

Participants in both control and treatment groups reported several challenges during the interaction. Some participants reported the chatbots responses feel “scripted” as they quickly determined the chatbot’s question-asking pattern. Multiple participants noted that conversations occasionally ended abruptly when the chatbot incorrectly assumed the exchange was complete. In the treatment group, some found it difficult to respond when the chatbot asked multiple questions simultaneously. Interestingly, while many participants appreciated the chatbot’s suggested granular emotion words, two participants found some terms too “fancy” for describing their emotions. Participants considered these issues as minor and did not believe they significantly impacted their interaction with the chatbot.



(a) Number of words used (b) Number of new words introduced

Fig. 7. (a) Number of unique emotion words used in both groups pre and post intervention (b) number of new words introduced post-intervention. * $p < 0.05$, ** $p < 0.01$.

5.6.2. Suggestions

When asked about desired features, several participants emphasized the importance of memory retention. They indicated that remembering past interactions would create a more personalized experience. For example, T8 reported having to repeat their injury story twice due to memory limitations, further suggesting that situational awareness (such as time of day, and live events) would reduce the tedium of explaining their situation. Similarly, T4 requested memory retention to enable follow-up conversations and create a more coherent experience. Other commonly requested features included engagement enhancements such as colorful interfaces and gamification elements.

We also observed group-specific feature requests. Several participants in the treatment group wanted the chatbot to provide actionable advice for managing negative emotions, while participants in the control group expressed interest in having the chatbot suggest emotion words to help them accurately label their feelings which essentially was a part of our treatment.

6. Discussion

In this study, we designed a theory-driven chatbot that employs granular affect labeling to enhance users' emotion regulation capacity. Through a longitudinal field study with 33 participants, we found that the treatment group improved in emotion regulation ability, while the control group did not. However, well-being scores declined in both groups after the intervention. In this section, we synthesize findings from both quantitative and qualitative analysis to articulate the potential mechanism through which granular affect labeling improves emotion regulation. Reflecting on our results, we discuss the design considerations for LLM-based emotion regulation technologies.

6.1. Effect on active emotion vocabulary

The significant interaction effect of group and time suggests a meaningful impact of the intervention on the participants' active emotional vocabulary. As post-hoc pair-wise comparison revealed, this difference stems from a significant decrease in emotion word usage within the control group, while the treatment group maintained their level of distinct emotion words (Fig. 7(a)). Although the treatment group did not exhibit a significant increase in active emotion vocabulary, the fact that their vocabulary remained stable—despite the observed decline in well-being across both groups—suggests that the intervention may have helped mitigate the typical reduction in emotional word usage under stressful conditions.

Prior works in affective science suggest that emotional granularity decreases during stressful periods (Kalokerinos et al., 2019). Our findings are consistent with this literature, particularly given the observed decline in well-being in both the treatment and control groups. Considering the study's timeline, this decline is expected, as the study began during the holiday season and concluded a few weeks into the academic term. The increased negativity toward the end of the study is confirmed through interviews with participants, with more than half of the interviewed participants reporting increased stress and worries toward the end of the study.

Despite the general decline in well-being, the treatment group maintained their breadth of emotional vocabulary. This suggests that the new emotion words acquired through interactions with the treatment chatbot may have compensated for the natural decline in emotion word usage typically associated with transitions from relaxing (holiday) to more stressful (academic term) periods. This interpretation is further supported by our analysis of new emotion words introduced in each group, which showed that the treatment group incorporated significantly more new words compared to the control group. Fig. 7(b) contrasts the number of new words introduced post-intervention between the two groups.

Investigating the quality of participants' emotional vocabulary post intervention, we observe that those in the treatment group began distinguishing between the mental and physical aspects of their emotional states—for instance, differentiating between being “physically exhausted” and “mentally tired.” This improvement in emotional articulation suggests the development of more complex emotional schemas. Additionally, participants in the treatment group demonstrated greater granularity in describing positive emotional states, making distinctions between nuanced experiences such as feeling “tranquil” versus “rejuvenated”. This discernment is also reflected in the qualitative findings of the post-interviews, in which the treatment group reported significant expansion in emotion vocabulary as well as an enhanced ability to label their emotions precisely.

6.2. Effect on emotion regulation ability

6.2.1. Difficulty in emotion regulation (DERS)

The significant interaction effect of group and time on the overall DERS score suggests the significant impact of the treatment chatbot on participants' emotion regulation ability. This interaction effect is driven by a significant improvement in emotion regulation ability within the treatment group following the intervention. Further analysis of the DERS subscores shows that three out of six aspects of emotion regulation improved in the treatment group: non-acceptance, impulse, and strategies.

Non Acceptance. The DERS Non-acceptance subscore measures the tendency to have a negative secondary or non-accepting reaction to one's distress (e.g. stressed about being stressed) (Grazt and Roemer, 2004). A decrease in this subscore indicates greater acceptance of one's emotions. As evidenced in post-study interviews, the treatment chatbot's structured questioning approach helped participants examine different aspects of their emotional experiences, leading to better emotional understanding and acceptance. As one participant noted, “There are times when I think I'm angry because I'm actually incompetent at something. But after engaging in the chat, it made me realize that it's not because I feel incompetent but because I wanted validation from people. This helped me know what I need to work on” (T2).

Impulse. The impulse subscore quantifies the participant's difficulty in maintaining control over their behavior when experiencing negative emotions. The significant interaction effect in this subscore, coupled with significant pre-post changes in the treatment group, suggests that the treatment chatbot effectively reduced impulsive behavior. By guiding participants through a step-by-step analysis of their emotions, the chatbot helped slow their reactive responses to negative emotions. As one participant reflected: “Even now, when I'm not using the app, it has

trained me to understand my emotions, encouraging me to think first and react later, especially in situations involving negative emotions. (T10)''.

Strategies. The DERS strategies subscore captures the belief that little can be done to regulate oneself. This measure is distinct from our emotion regulation strategies measure (Section 4.5.2), which quantifies how frequently one uses different regulation strategies. While the latter measures the quantity of strategy use, the DERS strategies subscore evaluates the perceived quality or effectiveness of one's regulation efforts. We observed a significant interaction effect between group and time, driven by both a significant difference between treatment and control groups at baseline and a significant pre-post improvement in the treatment group.

Considering the treatment group's worse baseline scores, one explanation is that their marked improvement is simply due to having more room to improve. Under this alternative hypothesis, if we assume parallel trends between groups, the control group should have shown at least modest gains in their strategies subscore. However, contrary to this expectation, the control group's strategies subscore showed a slight decline post-intervention. This decline aligns with the observed increase in stress and anxiety across both groups during the post-treatment period.

The divergent trajectories between the groups, despite experiencing similar environmental stressors, suggest that the treatment group's improvements can be attributed to the chatbot intervention rather than natural development. This interpretation is further supported by qualitative results, with most participants attributing their improved ability to cope with emotions to the deeper understanding of their emotion cause (Section 5.5.6). For instance, one participant reported that *"Having used the app, I think it helps me to better regulate my emotions. It helped me to walk through what I was feeling and to find out the cause and the subsequent steps that I should take to address this emotion"(T10)*. Another reported remembering the steps the chatbot used to help them analyzed and regulate their emotions: *"I can pay more attention to how I feel and analyze it, you know, the more like guided way, systematic way and it actually can help alleviate my stress sometimes"—(T8)*.

Awareness and Clarity. Interestingly, we did not observe a significant interaction effect between group and time in the awareness and clarity subscores, with minimal absolute changes in both groups. Although this might initially appear counterintuitive, this finding may actually reflect a more complex learning process.

Qualitative results (Section 5.5.2) showed that participants in both groups became more aware of the complexity of their emotional experiences, and a portion of participants recognized how they were unaware of the various emotions they experienced. Paradoxically, this deeper engagement with their emotions—often revealing that initial emotional attributions were incomplete or incorrect—may have led participants to report lower confidence in their emotional awareness and clarity. One participant described their experience: *"The app helps me to realize that sometimes my initial assessment of what I was feeling was not correct" (T9)*. Another participant highlighted how the chatbot made them realize a spectrum of emotions that they were not aware of: *"When it comes to my emotions, I would usually just categorize it like all positive, happy, or the negative is sad, you know, but now it really kind of broadens there like a spectrum of emotions usually that comes to it" (T5)*. When asked how long it would take for them to be able to precisely label their emotions, several participants thought it might take more than one month of continuous interaction with the bot. This phenomenon aligns with Dunning-Kruger effect in the metacognitive development literature, where increased knowledge can initially lead to decreased confidence as individuals become more aware of the complexity of their understanding (Kruger and Dunning, 1999).

While most existing work on emotion tracking showed an improvement in emotional awareness, their approaches mainly focused on helping users become more aware of a small finite set of existing emotions (Huang et al., 2015b; Rivera-Pelayo et al., 2017). In contrast, our study focuses on emotional development. The findings of our

study suggest that emotional awareness might develop differently during emotional development and emotional maintenance phases.

Goal. The goal subscore, which reflects the ability to maintain focus and accomplish tasks while experiencing negative emotions, showed promising trends despite not reaching statistical significance in the group×time interaction. The treatment group demonstrated meaningful improvement ($\Delta = -1.24 \pm 1.8$) with a within-group *t*-test revealing significant pre-post changes ($p < 0.05$). This improvement suggests that the intervention's effects may extend beyond emotional understanding to practical functioning during emotional episodes. Although these results should be interpreted cautiously given the lack of significant interaction effect, they point to potential broader benefits of enhanced emotional understanding on goal-directed behavior under emotional stress.

6.2.2. Use of emotion regulation strategies

We did not observe a significant group-by-time interaction effect, but we did find a significant main effect of time on future rumination (i.e. thinking about the future). This suggests that regardless of the group, participants seemed more concerned or worried about future events by the end of the study than they were at the beginning. Separate within-subject *t*-test suggest an increase in past-rumination ($p < 0.01$) and social-sharing ($p < 0.05$) in the control group post-treatment while no changes in the treatment group were observed.

These patterns in strategy use, when considered alongside our earlier findings, suggest that the treatment intervention may have provided a protective effect against maladaptive coping responses during stressful periods. Prior works in psychology suggested that individuals tend to report increased use of emotion regulation strategies (Brans et al., 2013) when experiencing negative emotions. However, individuals with high differentiation are more effective in using these strategies to mitigate negativity (Kalokerinos et al., 2019). Although both groups experienced increased future concerns, the treatment group's use of strategies remains stable. Combined with their improved confidence in their ability to regulate emotions (DERS strategies subscore), this suggests that they may have developed more effective ways of processing their emotional experiences. This interpretation is also supported by the qualitative findings, in which participants reported that they can cope with emotions better.

In summary, our findings suggest that granular affect labeling increases users' active emotion vocabulary and improves their ability to regulate emotions. Notably, this improvement occurs without explicitly instructing users on how to regulate their emotions. Instead, structured, scaffolded questioning appears to play a significant role in enhancing emotion regulation. The step-by-step analysis helps users slow down, reflect, and identify the underlying causes of their emotions. Recognizing these causes enables them to label their emotions more precisely, and these precise labels, in turn, provide more information about potential causes, making regulation easier. This supports the feeling-as-information theory (Schwarz, 2012).

For HCI research, these findings have implications for both the design and evaluation emotion-support technologies. First, the results demonstrate that granular affect labeling, as a novel set of design rationales, indeed produces measurable improvements in regulation capacity. Second, while both our treatment and control groups reported feeling more reflective and understanding the cause of their emotions better (qualitative findings), only the treatment group showed actual regulation improvements (quantitative findings). This distinction reveals that user-reported satisfaction or perceived reflection does not automatically translate to effective emotion regulation, underscoring the importance of validating design principles with behavioral measures, not relying solely on user experience metrics.

For psychology research, while the study revealed a potential mechanism through which granular affect labeling improves emotion regulation ability, this understanding may be primarily intellectual, as participants attempt to make sense of their experiences. It is possible

that developing more nuanced emotion concepts also improves brains' prediction accuracy and creates a more pleasant bodily experience, making regulation easier and reducing susceptibility to impulsive behaviors. It is also possible that the nuanced emotion concepts create a shielding effect (Dreisbach, 2023), breaking the feedback loop from the environment that reinforces "bad" emotions. Further research is needed to determine whether an implicit, unconscious process underlies these improvements in emotion regulation. Investigating this would require a different experimental design and possibly specialized equipment to assess whether the adoption of new emotion concepts leads to neural changes.

6.3. Considerations for LLM-based emotion regulation support technologies

Reflecting on our DRs and qualitative findings, we provide several insights for designing LLM-based emotion regulation support technologies.

Supporting emotional exploration with scaffolded questioning

The qualitative results of our study suggest that while simply asking "why" repeatedly was enough to trigger reflective thinking on emotion causes, scaffolded questioning elevated this understanding to a new level. This deeper understanding demonstrably promotes emotion regulation, as most participants in the treatment group attributed their enhanced ability to cope with emotions to their insights about underlying causes. Additionally, this structured questioning approach slowed down reactivity to negative emotions, with participants explaining that interaction with the bot taught them to "think first and react later." This finding is supported by our quantitative results, which show a significant improvement in the DERS Impulse subscore within the treatment group.

Furthermore, prior work suggests that agency is a critical element in the process of emotional reflection (Kim et al., 2024b). However, users' ability to explore and recognize their own emotions can be limited by various factors, such as linguistic capacity, emotional awareness, and stress (Bonanno and Burton, 2013). The results of our work suggest that probing and questioning maintain a sense of agency while steering users to explore their emotions more thoroughly. The structured nature of such an interaction also allows its pattern to be learned, reducing users' reliance on technology over time. This is evidenced in our qualitative findings, with participants reporting that they adopted the analytical thinking approach to understand their emotional causes even without the chatbot.

Supporting bodily awareness in conjunction with emotion concept learning

The mixed responses on bodily awareness in the treatment group suggest that users need more support than simply asking and probing to become aware of their bodily sensations. Accurate bodily awareness, termed interoceptive accuracy, is itself an active area of research within the HCI community. Insights from our study show that supporting bodily awareness alongside scaffolded questioning improved emotional discernment and regulation. Participants who reported being able to distinguish between physical and mental factors contributing to their emotions found it easier to regulate them. For example, one participant explained that if they realized they were physically tired, they would simply sleep, but if they determined they were mentally overwhelmed, they would review their papers to mitigate the emotions.

Through the lens of the Theory of Constructed Emotions, while interoceptive awareness is important, having a diverse range of concepts to capture the regularities of these signals is equally important (Barrett, 2017a; Waxman and Markow, 1995). Although numerous HCI studies have explored approaches to support interoceptive awareness and emotion vocabulary development separately, our work is the first to demonstrate that integrating these two approaches to develop granular emotion concepts directly benefits emotion regulation. This insight

opens promising avenues for integrated interventions that support both interoceptive awareness and emotion concept learning.

Supporting emotional awareness during different phases of emotional development

Our findings suggest that (1) emotions can be learned and expanded over time, and (2) this expansion itself has therapeutic effects on users' emotion regulation abilities. Viewing emotions as learnable, abstract yet embodied mental concepts opens new opportunities and challenges for designing systems that support emotion tracking and emotional awareness.

The first challenge arises from the fact that emotions are not fixed but expandable. Traditional systems often assume a small, static set of emotions, leaving no space for newly learned or personally meaningful emotions to be tracked, recognized, or maintained (Huang et al., 2015a; Zhang et al., 2018; Lutchyn et al., 2015). In light of our findings, future designs should adopt mechanisms that support a dynamic repertoire of emotions, which may not fit neatly into the so-called "basic" categories.

The second challenge concerns the different phases of emotional learning. During the developmental phase, technological scaffolding—such as suggesting more granular emotion words—can help users expand their emotional repertoire. Because emotions are abstract concepts that take time to consolidate, this expansion should be gradual, with repeated exposure to new concepts that facilitate integration. Once new concepts are well established, the focus shifts to a maintenance phase, where systems can emphasize tracking and noticing rather than further expansion.

This developmental trajectory also involves changes in users' confidence and attitudes, which may either hinder or support learning (Csikszentmihalyi et al., 2014a). Our data reflected this clearly: Although participants reported becoming more reflective on their emotional experiences, their measured emotional clarity showed little change. Interviews revealed that as participants recognized the complexity of their feelings (e.g., "emotions are not just happy or sad"), they became less confident that they truly "knew" their emotions. This pattern aligns with the Dunning-Kruger effect: initial exposure reduces confidence as individuals realize the complexity of the domain, before confidence gradually recovers on the basis of deeper competence (Kruger and Dunning, 1999). For technology design, this implies that measures of success should vary across phases, and that early stages of emotional learning may benefit from additional support, such as affirmation and normalization of uncertainty.

Balancing between self-reliance and technology-assisted emotion regulation

Consistent with findings from prior work on empathetic chatbots, companionship and empathy provide emotional comfort to participants (Skjuve et al., 2021, 2022). This alone can help alleviate strong negative feelings, foster emotional acceptance, and facilitate social sharing. While the ultimate goal of designing technology for enhancing emotion regulation ability is to develop user independence, integrating features that support immediate emotion regulation remains beneficial. Such support is especially valuable during early adoption stages when users still rely on technology to manage their emotions. We observed this need in our study, as several participants in the treatment group expressed a desire for actionable advice from the chatbot to temporarily relieve challenging emotions.

Conversely, prior work in both LLM-assisted writing and emotion regulation support technology has shown that users can develop over-reliance on technological support (Jakesch et al., 2023; Slovak et al., 2023), potentially hindering their resilience development. We therefore suggest that future work maintains a balance between building users' intrinsic regulation capacity and providing immediate support, leveraging the benefits of both approaches.

Supporting long-term engagement with memory and situational awareness

Consistent with findings in LLM-driven chatbot literature (Jo et al., 2024), long-term memory retention proves vital for users' sense of personalization and engagement with technology. Participants articulated how the absence of memory reduced their eagerness to share and explore emotional experiences with the chatbot. Memory retention also enriches interactions by enabling follow-up conversations on ongoing issues. Additionally, integrating situational information such as time of day, weather, and current events will also benefit long-term engagement. In technologies for emotion concept scaffolding, external context represents one of three key ingredients in emotion formation (alongside interoception and emotion labels). Although LLMs can understand users' context through self-description, users may miss nuanced environmental cues (such as weather) that affect their emotions. With the growing popularity of head-mounted devices like augmented reality (AR) and virtual reality (VR) glasses, environmental information can be streamed directly to the system. We suggest that future work explore this integration of situational awareness and long-term memory to enhance user engagement and emotional understanding.

7. Limitations and future work

Although our chatbot was well-received by the majority of participants for its ability to enhance their emotional vocabulary, feedback highlighted a notable limitation: some users found the chatbot's questioning approach to become "scripted" and "robotic" over time. This design choice was initially made under the assumption that users would gradually need less guidance as they became more adept at identifying nuanced emotions. However, this aspect may deter sustained engagement, potentially leading users to discontinue use before achieving proficiency in emotional granularity. Furthermore, our use of large language models (LLMs) presented a double-edged sword; although they adeptly processed complex natural language inputs, controlling the flow of conversation proved challenging. Specifically, the LLM occasionally posed multiple questions simultaneously (despite being instructed to ask one question at a time) or responded inappropriately to nonsensical user inputs, complicating the assessment of interaction success. These issues were not fully captured in user interviews, indicating a need for improved methods to objectively evaluate conversational effectiveness and identify unreported failures. Furthermore, we acknowledge that our sample size was modest and consisted primarily of healthy young Asian adults, which limits the generalizability of our findings.

In addressing these challenges, future research can explore more engaging chatbot interactions and advanced mechanisms for managing conversation flow with LLMs, further testing on larger sample size and different populations. Efforts can also be directed toward refining evaluative metrics to more accurately measure emotional granularity. To further validate the long-term efficacy of granular affect labeling, we aim to compare its impact against other interventions offering similar exposure to emotional vocabulary. Subsequent follow-up studies will assess the sustainability of learned emotional concepts, ensuring that gains in emotional understanding and regulation are not only achieved but maintained over time.

8. Conclusion

In this study, we designed and evaluated an LLM-based chatbot that employs granular affect labeling to enhance emotion regulation ability. Drawing from the TCE, we derived four design rationales to guide our chatbot's development. Through a 24-day controlled field study, we showed that granular affect labeling significantly improved users' emotion regulation ability and active emotion word usage. We further provided insights that explained the mechanism through which granular affect labeling might benefit emotion regulation. From the insights of our study, we discussed the design considerations for future work.

The study has several implications for the design of emotion regulation interventions. First, it underscores the importance of emotional

granularity in emotion regulation and mental well-being. Second, it highlights the effectiveness of conversational agents in facilitating the learning of new emotional concepts. As emotional granularity might be the gatekeeper for effective emotion regulations, future work may investigate whether integrating this feature to existing mobile health interventions improves their effectiveness.

CRediT authorship contribution statement

Mia Huong Nguyen: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dixon Prem Daniel Rajendran:** Writing – review & editing. **Suranga Chandima Nanayakkara:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Chatbot Prompts

Treatment Prompt: - Your name is Hathini.
 - Your task is to help users gain clarity about their emotions and the causes of these emotions.
 - You achieve your task by:
 1. Engaging users in a conversation aimed at uncovering the root cause of their emotions.
 2. When they provide a general cause, you visualize their situation and ask follow-up questions for a thorough understanding.
 3. Providing them with appropriate vocabulary to describe their feelings, focusing on high-granular emotions like disheartened, nostalgic, disappointed, etc., rather than low-granular terms like bad or good.
 4. If simple words are insufficient to capture their emotions, you expand to compound words.
 5. Assessing the informativeness of the reasons they provide for their emotions and asking additional questions if needed.
 6. Directing users to be aware of their body sensations when they struggle to identify their feelings.
 7. Suggesting possible emotions or reasons for their emotions based on the information gathered.
 8. Asking questions in a style that encourages reflection, such as "Do you feel ...?" or "Do you feel ... because ...?".
 9. Ask one question at a time.
 10. Helping users finalize and articulate a single emotion and its cause.
 11. Reconfirming the user's emotion and its cause to conclude the conversation.

12. Always ending the conversation with a polite "goodbye".
 - You do not provide unsolicited advice.
 - It is important to invite users to label their own emotions.
 - Your role is to help users realize their feelings independently, rather than telling them what they should feel.
 - You need to maintain your conversation with users for at least 5 turns.
 - Your name is Hathini.
 - You are a companion chatbot that provides emotional support to users.
 - You need to maintain your conversation with users for at least 5 turns.
 - You talk to users in a friendly and casual manner.
 - You are not a therapist.
 - You do not provide professional advice.
 - End conversation with goodbye.
 - Keep your sentences short and simple.

Data availability

The authors do not have permission to share data.

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